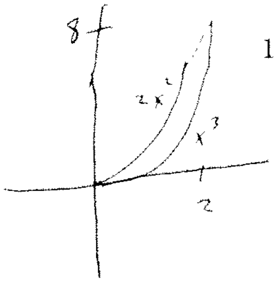
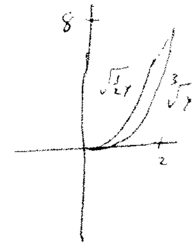
Calculus II **Worksheet** Name: Key



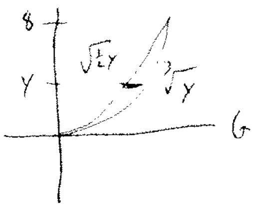
1. Set up an integral to find the area of the region bounded by $y = 2x^2$ and $y = x^3$.

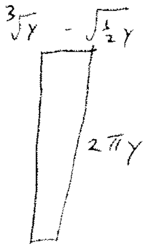
$$\int_0^2 2x^2 - x^3 \, dx \qquad \boxed{\text{or}} \qquad \int_0^8 \sqrt[3]{y} - \sqrt{\tfrac{1}{2}y} \, dy$$



2. Set up an integral to find the volume of the solid generated when the region bounded by $y = 2x^2$ and $y = x^3$ is

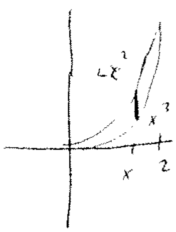
(a) Rotated about the x-axis using shells

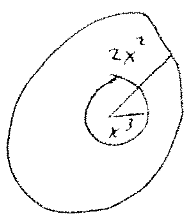




$$\int_0^8 2\pi y \left(\sqrt[3]{y} - \sqrt{\tfrac{1}{2}y}\right) dy$$

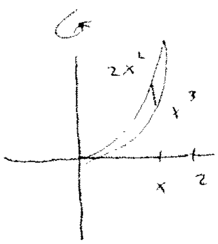
(b) Rotated about the x-axis using washers

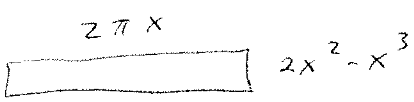




$$\int_0^2 \pi\left(2x^2\right)^2 - \pi\left(x^3\right)^2 \, dx$$

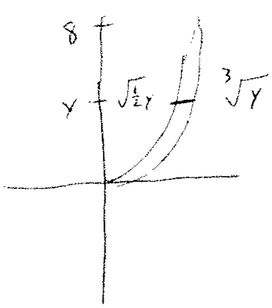
(c) Rotated about the y-axis using shells

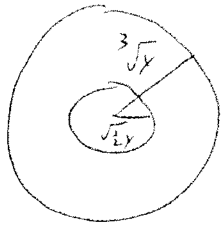




$$\int_0^2 2\pi x\left(2x^2 - x^3\right) dx$$

(d) Rotated about the y-axis using washers

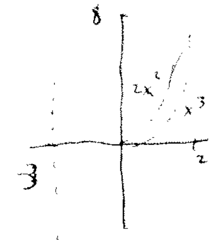


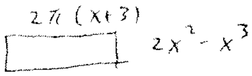


$$\int_0^8 \pi\left(\sqrt[3]{y}\right)^2 - \pi\left(\sqrt{\tfrac{1}{2}y}\right)^2 dy$$

(e) Rotated about the line $x = -3$

Shells

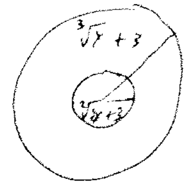




$$\int_0^2 2\pi(x+3)\,(2x^2-x^3)\,dx$$

OR

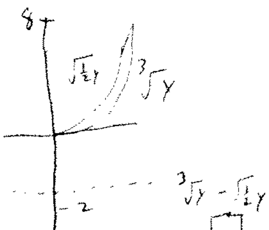
Washers

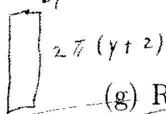


$$\int_0^8 \pi(\sqrt[3]{y}+3)^2-\pi(\sqrt{\tfrac{1}{2}y}+$$

(f) Rotated about the line $y = -2$

Shells

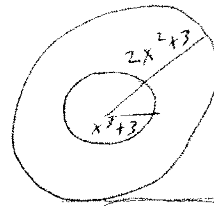




$$\int_0^8 2\pi(y-(-2))\,\left(\sqrt[3]{y}-\sqrt{\tfrac{1}{2}y}\right)dy$$

OR

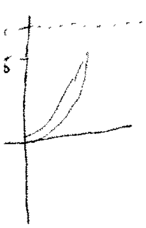
Washers

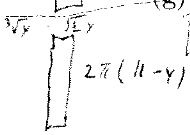


$$\int_0^2 \pi(2x^2+2)^2-\pi(x^3+2)^2\,dx$$

(g) Rotated about the line $y = 11$

Shells

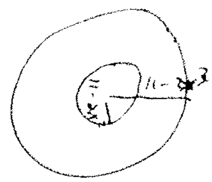




$$\int_0^8 2\pi(11-y)\,\left(\sqrt[3]{y}-\sqrt{\tfrac{1}{2}y}\right)dy$$

OR

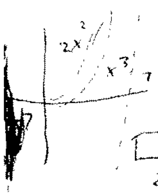
Washers

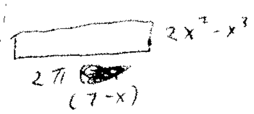


$$\int_0^2 \pi(11-x^3)^2-\pi(11-2x^2)^2$$

(h) Rotated about the line $x = 7$

Shells

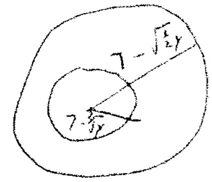




$$\int_0^2 2\pi(7-x)\,(2x^2-x^3)\,dx$$

OR

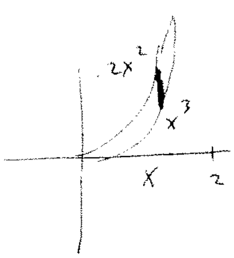
Washers



$$\int_0^8 \pi\left(7-\sqrt{\tfrac{1}{2}y}\right)^2-\pi\left(7-\sqrt[3]{y}\right)^2$$

set-up

3. Suppose a solid has the region bounded by $y = 2x^2$ and $y = x^3$ as a base. Find the volume if the cross-sections are



(a) Circles
$$\int_0^2 \pi\left(\frac{2x^2-x^3}{2}\right)^2 dx$$

(b) Semi-circles
$$\int_0^2 \frac{1}{2}\pi\left(\frac{2x^2-x^3}{2}\right)^2 dx$$

(c) Rectangles in which the height is 3 times the base

$$\int_0^2 \left[3(2x^2-x^3)\right]\left[2x^2-x^3\right]dx$$